Roll No.: ....

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B.Tech | Semester: I

Name of the Course: Introduction to Python Programming | Course Code: TCS 102

Time: 3 Hours | **Maximum Marks: 100**

Note:

i. **All questions are compulsory.**
ii. Answer any two sub questions among a, b & c in each main questions
iii. Total marks in each main question are twenty.
iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Describe the block diagram of a computer system and explain how its key components—Input Unit, Output Unit, Memory Unit, and the Central Processing Unit function.	CO1
(b)	Discuss what a computer network is and what their types are. Mention at least five important Internet services used in daily life.	CO1
(c)	Draw a flowchart and develop a python program for a ticket booking system for a cinema hall that takes age as an input from the user and prints the final ticket price. The standard price of a ticket is 150. The theatre gives discount based on the age. If the age is below 12 the discounted price is 100, if the age is greater than 60 the discounted price is 60. Also check if the age is below 0 or greater than 120 print "Invalid age".	CO2
Q2		
(a)	Distinguish between lists, tuples, dictionary and sets based on the following parameters: i. Ordered or unordered ii. Mutability iii. Syntax iv. Indexing v. Duplication	CO4
(b)	Implement a menu driven python program for marks management system that uses dictionary for storing the student's marks where student name is a key and marks is a value. The system performs the following operations based on the user choice: i. Input a new student name and marks from the user and add it into a dictionary. ii. Update the marks of the student already present in the dictionary. Both name and marks should be taken as a user input. iii. Display the marks of the student as input by the user. iv. The program should handle the valueError exception when user gives invalid input like integer in place of string or vice-versa and generic exceptions for any other exception raised.	CO2
(c)	i. Explain the concept of lambda functions in Python with a help of code snippet and discuss when they are typically used. ii. Develop a custom python module named list_util.py that has the following functions: 1. find_max(list): a function to find maximum in a list. 2. find_min(list): a function to find minimum in a list. 3. find_sum(list): a function to find sum of the list. The main.py file should import the custom module and call the functions.	CO3

Q3		
(a)	i. Compare the difference between search(), and match() functions of re module with the help of example of each. ii. Build a python program that reads a text file and replaces all the vowels present in it with star symbol(*) using regular expressions and write in a same file.	CO3 CO4
(b)	Describe the following with the help of code snippet: i. read() ii. readline() iii. readlines() iv. write() v. writelines()	CO4
(c)	Create a python program that reads a file bank.txt that contains the cust_id, cust_name, acc_type, balance and performs the following operations: i. Find the customer names with balance greater than 50000. ii. Find the customer ids where account type "saving". iii. Find the customer names starting with A. iv. Calculate the total balance of all the customers. v. Update the acc_type as "current" where cust_id is 105. The sample data stored in a file is as follows: cust_id cust_name acc_type balance 101 Ajay current 60000 102 Naman saving 30000	
Q4		
(a)	Elaborate polymorphism in object oriented programming. Identify how function overloading differs from function overriding with a help of a python code.	CO5
(b)	Explain how private protected and public access specifiers ensures data hiding. Demonstrate encapsulation with the help of a python class Student that has name as public member, age as protected member and grade as private member and show_info() a public function that displays all the members.	CO5
(c)	Describe the following types of inheritance with the help of an example: i. Single inheritance ii. Multi-level inheritance iii. Hierarchical inheritance iv. Multiple inheritance v. Hybrid inheritance	CO5
Q5		
(a)	Analyze the importance of data cleaning in data preprocessing. List 2 preprocessing techniques done for data cleaning. Mention 5 techniques used for Exploratory Data Analysis.	CO6
(b)	A teacher wants to visualize the marks of students in three subjects: Math, Science, and English. The students name and marks in each subject are given in individual numpy arrays. Build a Python program using Matplotlib or seaborn to perform the following: i. Display a bar graph of marks in Math subject with bar color as red. ii. Plot a histogram showing the distribution of marks in Science with bins = 5. iii. Show a pie chart for marks in English with students as label. iv. Create a scatter plot between marks of Math and Science. v. Draw a line chart to show the trend in marks of English among students using the marker '*'. The charts should have labeled x-axis y-axis and title.	CO6
(c)	A retail company maintains a dataset named sales_data.csv that contains the following columns: prod_id, category, price, region, revenue. As a data analyst perform the following tasks: 1. Read the data from csv into a dataframe. 2. Display the first and last 5 rows. 3. Find the number of null values. 4. Show the structural info and statistical info of the dataframe. 5. Impute the missing values in revenue column with the mean revenue. 6. Find the no. of unique product categories. 7. Find all the products sold in 'North ' region 8. Update the price of the product as 2500 having prod_id as 1001.	CO6